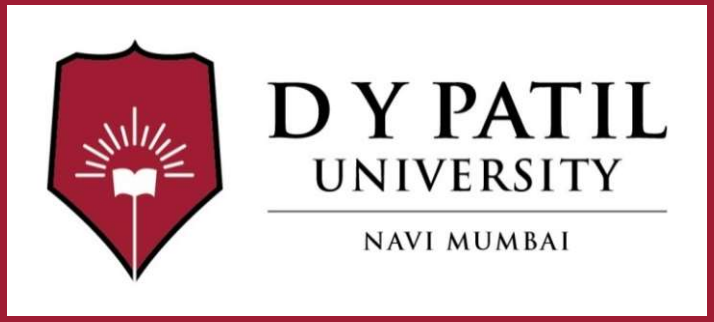


Embedded Systems

Module 1: Introduction to Embedded systems:

Index
Week-1

Lecture 7 – Embedded Systems and general-purpose computer systems	3
Lecture 8 – classifications, applications and purpose of embedded systems	11
Lecture9– Features of Embedded cores Microcontroller, ASIC, RISC and CISC cores	16
Lecture 10 – Comparison of microprocessor and microcontroller	28

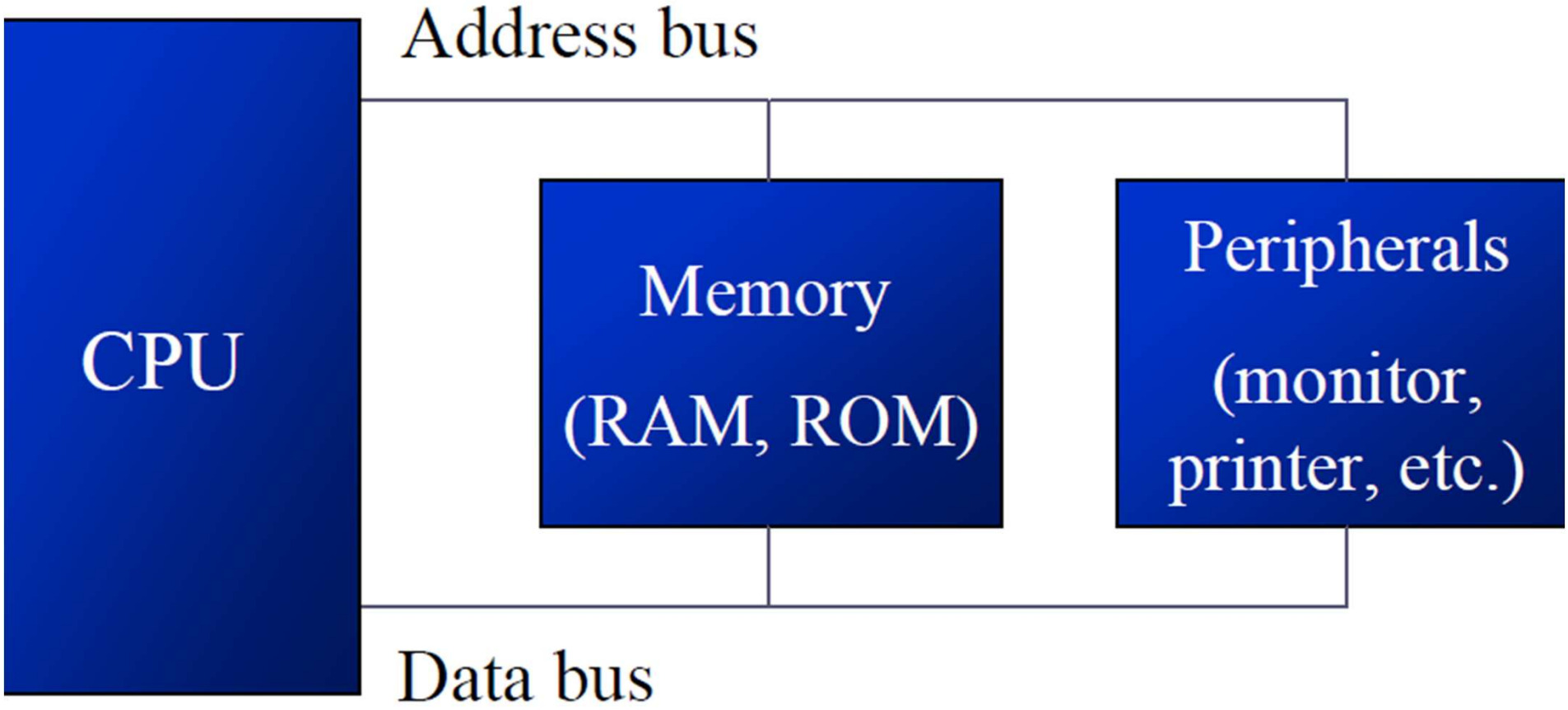
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Lecture 7

Embedded Systems and General-Purpose Computer Systems



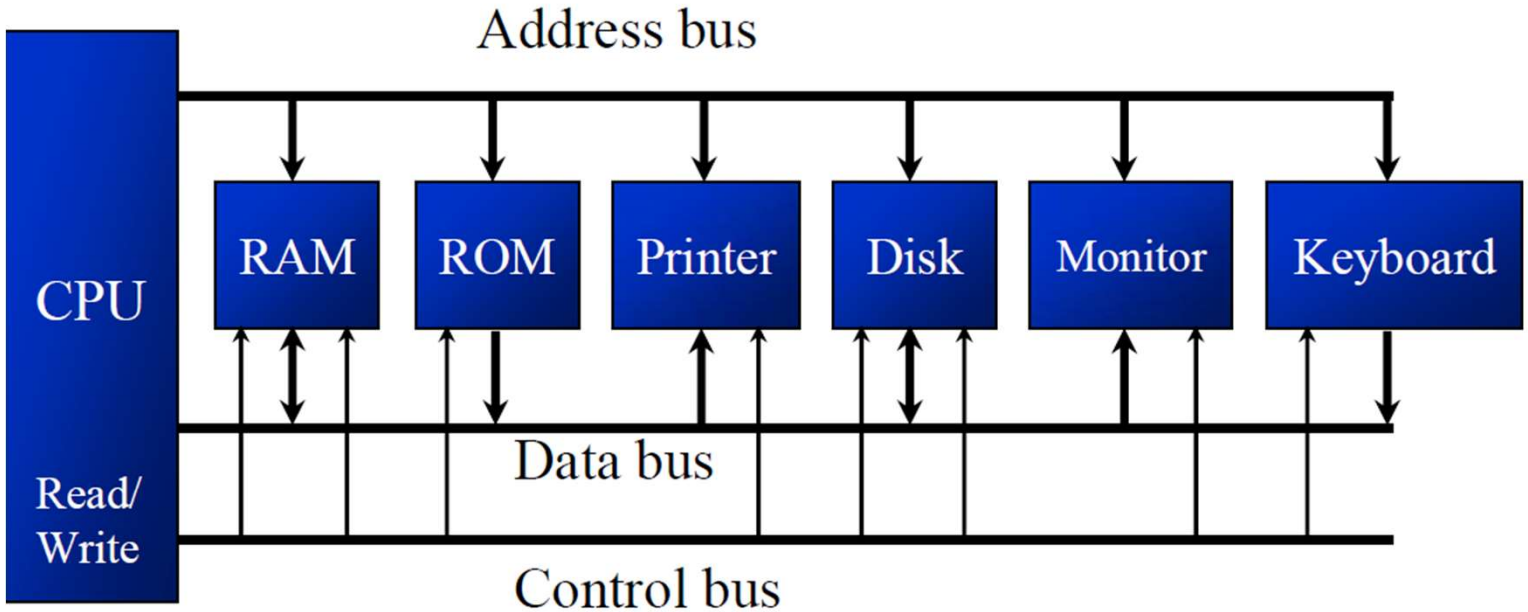
Internal Organization of Computer



Internal Organization of Computer

The CPU is connected to memory and I/O through strips of wire called a bus
Carries information from place to place.

- Address bus
- Data bus
- Control bus



Internal Organization of Computer

Address bus

- For a device (memory or I/O) to be recognized by the CPU, it must be assigned an address
- The address assigned to a given device must be unique
- The CPU puts the address on the address bus, and the decoding circuitry finds the device

Data bus

- The CPU either gets data from the device or sends data to it

Control bus

- Provides read or write signals to the device to indicate if the CPU is asking for information or sending it information

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Data Buses

The more data buses available, the better the CPU

- Think of data buses as highway lanes
- More data buses mean a more expensive CPU and computer
- The average size of data buses in CPUs varies between 8 and 64
- Data buses are bidirectional
- To receive or send data
- The processing power of a computer is related to the size of its buses

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Address Buses

- The more address buses available, the larger the number of devices that can be addressed
- The number of locations with which a CPU can communicate is always equal to 2^x, where x is the address lines, regardless of the size of the data bus.
- ex. a CPU with 24 address lines can provide a total of 2^{24} or 16M bytes of addressable memory
- Each location can have a maximum of 1 byte of data, since all general-purpose CPUs are byte addressable
- The address bus is unidirectional

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CPU Relation to RAM and ROM

- For the CPU to process information, the data must be stored in RAM or ROM, which are referred to as primary memory
- ROM (Non Volatile) provides information that is fixed and permanent
  - Tables or initialization program
- RAM (Volatile) stores information that is not permanent and can change with time
  - Various versions of OS and application packages
  - CPU gets information to be processed first form RAM (or ROM)
  - if it is not there, then seeks it from a mass storage device, called secondary memory, and transfers the information to RAM

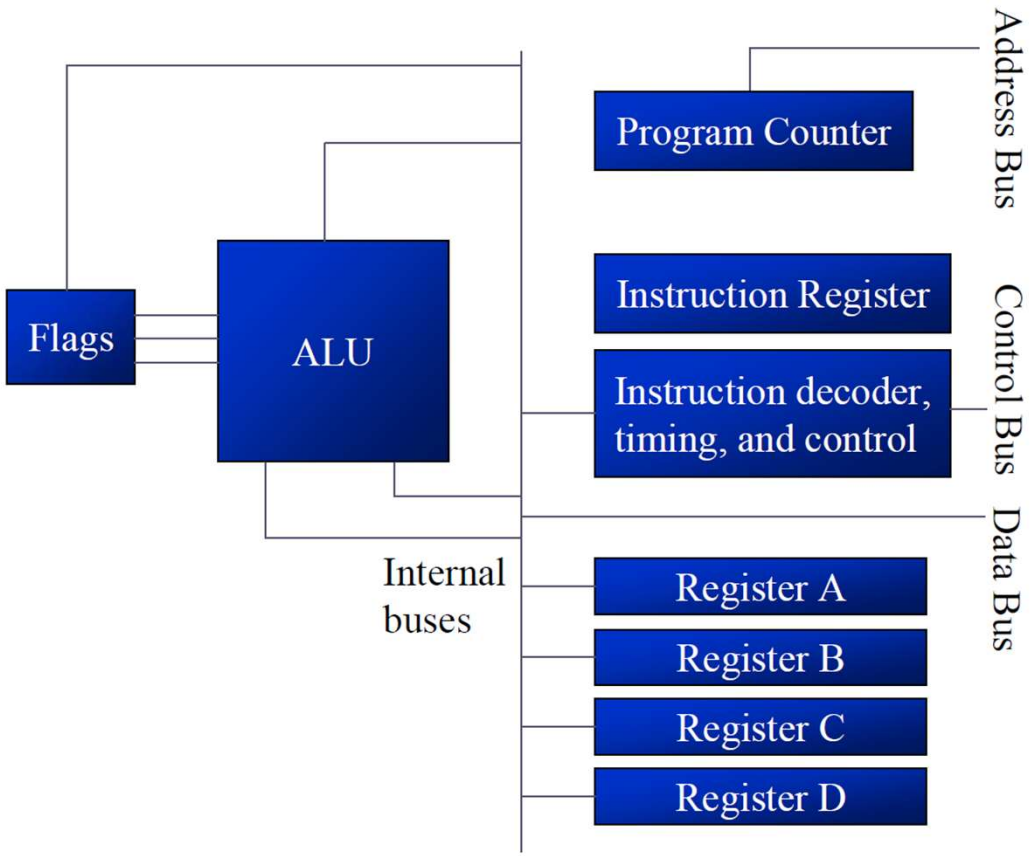
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Registers

- The CPU uses registers to store information temporarily
  - Values to be processed
  - Address of value to be fetched from memory
- In general, the more and bigger the registers, the better the CPU
  - Registers can be 8-, 16-, 32-, or 64-bit
  - The disadvantage of more and bigger registers is the increased cost of such a CPU

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Internal Organization of CPU



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Internal Organization of CPU

ALU (arithmetic/logic unit)

- Performs arithmetic functions such as add, subtract, multiply, and divide, and logic functions such as AND, OR, and NOT

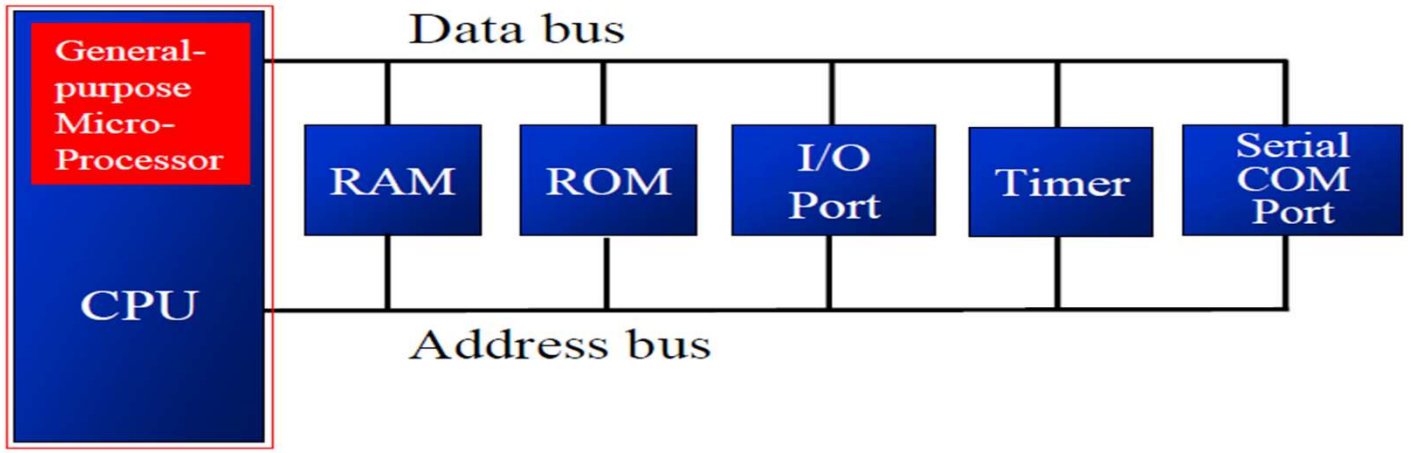
Program counter

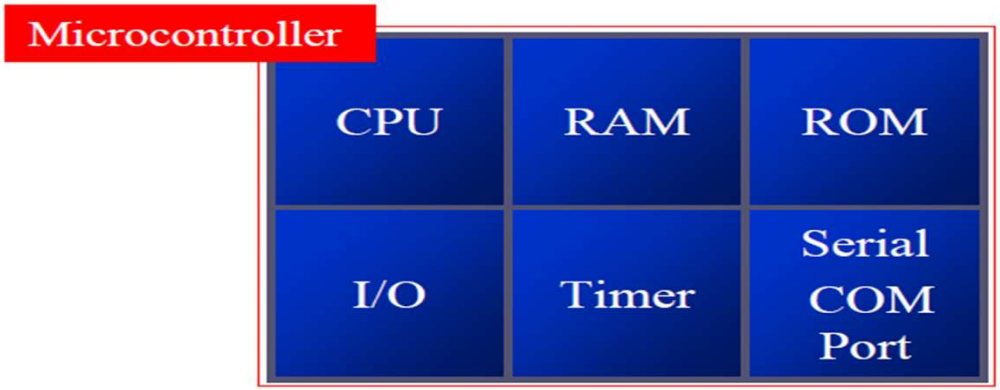
- Points to the address of the next instruction to be executed
- As each instruction is executed, the program counter is incremented to point to the address of the next instruction to be executed

Instruction decoder

- Interprets the instruction fetched into the CPU
- A CPU capable of understanding more instructions requires more transistors to design

Microcontroller Vs General Purpose Microprocessor





Microcontroller Vs General Purpose Microprocessor

General-purpose microprocessors contains

- No RAM
- No ROM
- No I/O ports

Microcontroller has

- CPU (microprocessor)
- RAM
- ROM
- I/O ports
- Timer
- ADC and other peripherals

Microcontroller Vs General Purpose Microprocessor

General-purpose microprocessors

- Must add RAM, ROM, I/O ports, and timers externally to make them functional
- Make the system bulkier and much more expensive
- Have the advantage of versatility on the amount of RAM, ROM, and I/O ports

Microcontroller

- The fixed amount of on-chip ROM, RAM, and number of I/O ports makes them ideal for many applications in which cost and space are critical
- In many applications, the space it takes, the power it consumes, and the price per
- unit are much more critical considerations than the computing power

MICROCONTROLLERS AND EMBEDDED PROCESSORS

An embedded product uses a microprocessor (or microcontroller) to do one task and one task only. There is only one application software that is typically burned into ROM.

A PC, in contrast with the embedded system, can be used for any number of Applications.

- It has RAM memory and an operating system that loads a variety of applications into RAM and lets the CPU run them
- A PC contains or is connected to various embedded products
- Each one peripheral has a microcontroller inside
- it that performs only one task

Embedded System

Embedded systems and general-purpose computers are both types of computer systems, but they serve different purposes. Embedded systems are designed for specific tasks within larger devices, while general-purpose computers are designed for a wide range of tasks and user interaction.

Designed to perform specific tasks within a larger system, like a washing machine, car, or medical device.

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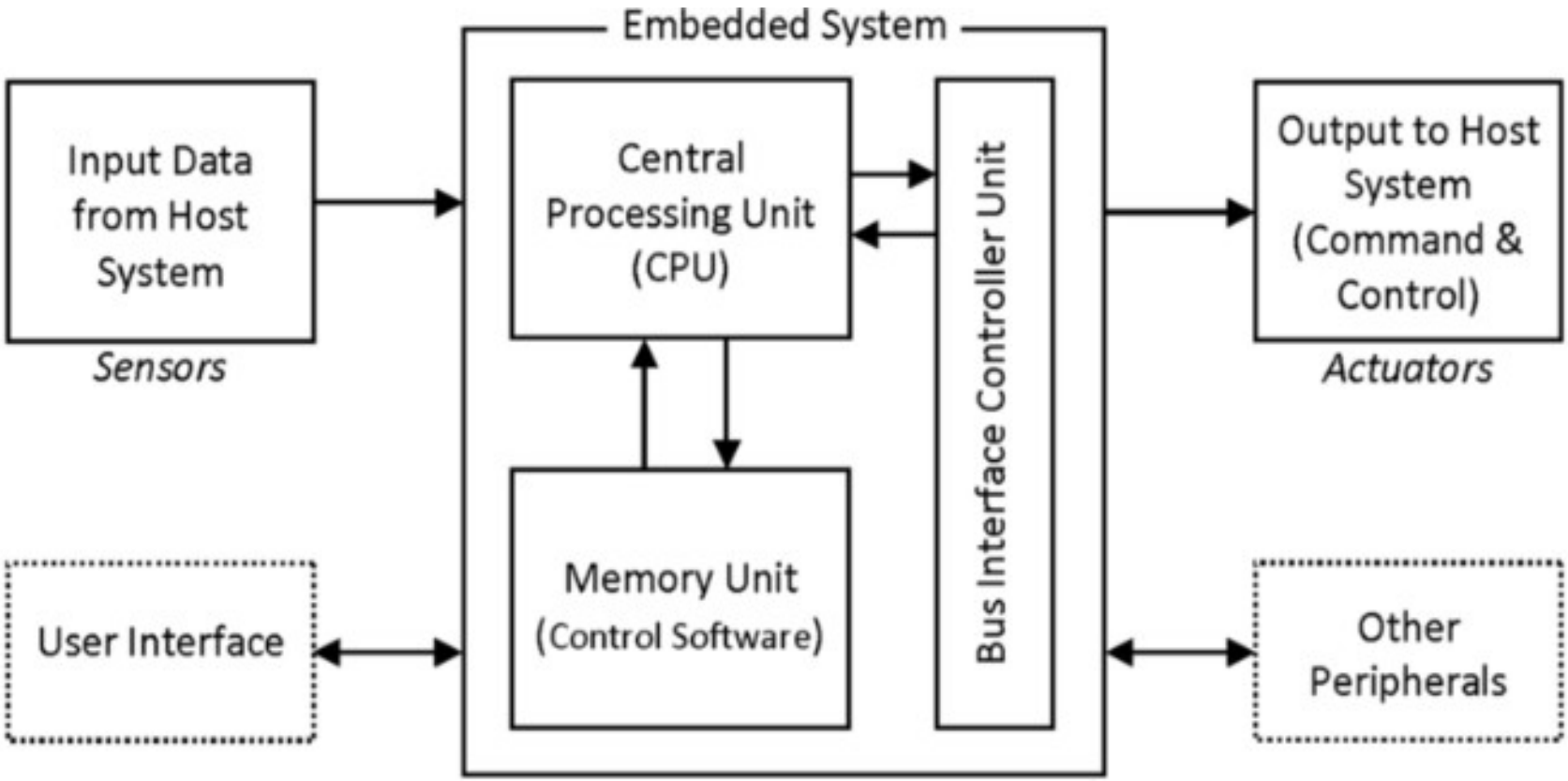
Embedded system

Characteristics:

- **Dedicated Function:** Optimized for a single or limited set of functions.
- **Hardware and Software Integration:** Combine hardware and software components to perform a specific task.
- **Resource Constraints:** Often have limited memory, processing power, and energy consumption compared to general-purpose computers.
- **Real-time Constraints:** Some embedded systems must operate within strict time constraints (e.g., a pacemaker).
- **Reliability:** Designed for high reliability and often operate without user intervention.
- **Examples:** Microcontrollers in appliances, engine control units in cars, and sensors in medical devices.

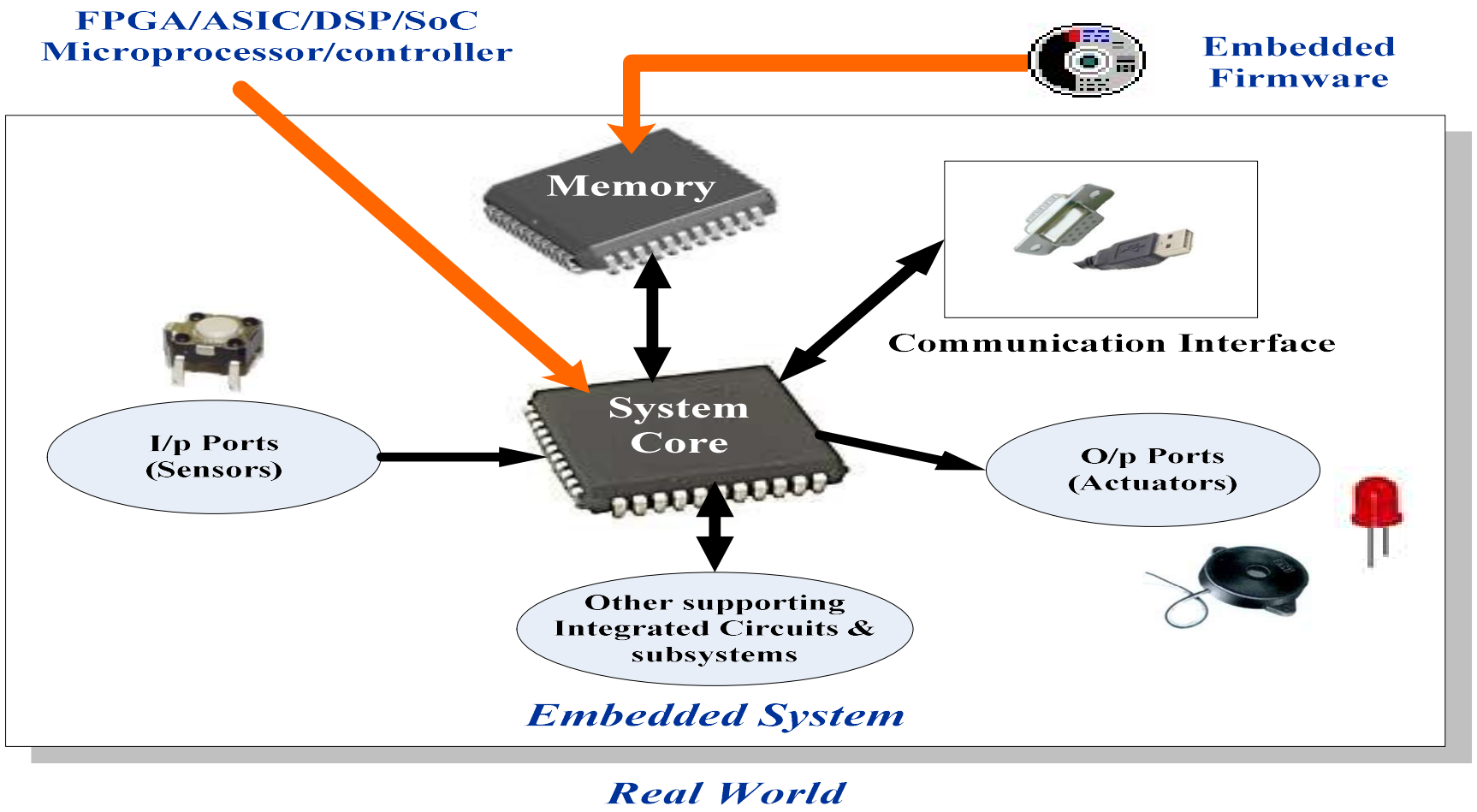
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Embedded system



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Embedded system



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The Core of the Embedded Systems

The core of the embedded system falls into any one of the following categories.

- **General Purpose and Domain Specific Processors**
  - Microprocessors
  - Microcontrollers
  - Digital Signal Processors
- **Programmable Logic Devices (PLDs)**
- **Application Specific Integrated Circuits (ASICs)**
- **Commercial off the shelf Components (COTS)**

Lecture 8

Classifications, Applications And Purpose Of Embedded Systems



Classifications of Embedded Systems: Based on Performance and Functional Requirements

- **Real-Time Embedded Systems**: Designed to process data and provide outputs within strict time constraints.
  - *Hard Real-Time Systems*: Missing a deadline can lead to system failure.
  - *Soft Real-Time Systems*: Deadlines are important but not critical.
  - *Examples*: Airbag systems, medical monitoring devices.
- **Stand-Alone Embedded Systems**: Operate independently without the need for a host system.
  - *Examples*: Digital cameras, MP3 players, microwave
- **Networked Embedded Systems**: Connected to a network to access resources and communicate with other systems.
  - *Examples: Home security systems, ATMs, point-of-sale terminals.*
- **Mobile Embedded Systems**: Portable devices with embedded systems.

  Examples: Smartphones, wearable fitness trackers, digital cameras.

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Classifications of Embedded Systems: Based on Microcontroller Performance

- **Small-Scale Embedded Systems:** Utilize 8-bit or 16 bit microcontrollers with limited memory and processing power.
  - *Examples*: Simple home appliances, basic toys.

- **Medium-Scale Embedded Systems:** Employ 16-bit or 32-bit microcontrollers, offering moderate complexity.
  - *Examples*: Washing machines, printers, automotive systems

- **Sophisticated Embedded Systems:**Use 32-bit or higher microcontrollers or digital signal processors (DSPs), supporting complex functionalities.
  - *Examples*: Advanced medical equipment, aerospace systems, industrial automation.

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Applications of Embedded Systems

Embedded systems are pervasive across various sectors:

- Consumer Electronics: Smart TVs, digital cameras, gaming consoles.
- Automotive Industry: Engine control units, anti-lock braking systems, infotainment systems.
- Healthcare: Medical devices like pacemakers, insulin pumps, imaging systems. Industrial Automation: Robotic arms, assembly line controllers, process monitoring systems.
- Telecommunications: Routers, switches, mobile phones.
- Automation: Smart thermostats, security systems, lighting controls

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Applications of Embedded Systems

Purpose of Embedded Systems

The primary objectives of embedded systems include:

- **Dedicated Functionality**: Perform specific tasks efficiently within a larger system.
- **Real-Time Operation**: Provide timely responses to inputs, crucial in applications like medical monitoring and automotive safety.
- **Resource Efficiency**: Optimize power consumption, processing power, and memory usage.
- **Reliability and Stability**: Operate continuously with minimal human intervention, often in critical applications.
- **Cost-Effectiveness**: Reduce overall system costs by integrating specific functionalities into dedicated hardware and software components.

Summary: Embedded systems are specialized computing units designed to perform dedicated functions within larger systems. Their classifications are based on performance, functionality, and hardware capabilities, and they are integral to various applications across industries due to their efficiency, reliability, and real-time processing capabilities.

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Lecture 9

Features Of Embedded Cores Microcontroller, ASIC, RISC And CISC Cores



Features of Embedded cores

1. **CPU (Processing Core)**

   Executes arithmetic, logic, and control tasks. Common architectures include 8-, 16-,32-bit cores—e.g., ARM Cortex-M0/M3/M4, AVR, PIC .Some include optional extensions like floating-point units (FPU), bit-banding, memory protection units (MPUs), and security features

2. **Memory System**
   - Flash/ROM for firmware; RAM for runtime data; often EEPROM for non-volatile settings .
   - Upper-end MCUs may include tightly-coupled memory (TCM) for low-latency operations

3. **Clock & Power Management**
   - Internal/external oscillators set clock signals; low-power modes (sleep/deep sleep) help conserve . Fine-grained power control (e.g., freezing some peripherals while the core sleeps).

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Features of Embedded cores: Peripheral & I/O Features

4. **GPIO (General-Purpose I/O)**
   - Multi-mode digital/analog pins (input/output, open-drain, pull-ups), anywhere from a few to dozens
5. **Timers, Counters & PWM Modules**
   - Essential for timekeeping, delays, measurement, PWM output for motor, LED, and servo control
6. **Analog Modules (ADC/DAC/Comparators)**
   - ADCs convert sensor outputs to digital; DACs generate analog signals; comparators assist threshold detection
7. **Communication Interfaces**
   - Serial: UART, SPI, I^2C
   - Advanced: CAN, Ethernet, USB, sometimes Zigbee, Bluetooth; commonly multiple interfaces onboard blog.
8. **Interrupt Controller**
   - Nested vector interrupts with priority support; able to wake CPU from low-power on events

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Why These Features Matter ?

- **Real-time control:** Precision timers, interrupts, and deterministic CPU behavior are vital.
- **Low power:** Energy-saving modes and efficient clocking are essential for battery-powered systems.Robust interfacing: ADCs, communication buses, and GPIOs enable interaction with sensors, motors, UI elements, other MCUs.
- **Security:** In IoT and connected devices, hardware isolation and crypto support are increasingly mandatory.Flexibility:
- **Programmable peripherals** (PIO, CPLD-like blocks) let designers customize functionality in silicon.

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Application-Specific Integrated Circuits (ASIC)

ASICs (Application-Specific Integrated Circuits): **Application-Specific Integrated Circuit**. It is a **custom-designed chip** made for a **specific task**.

Example: A chip inside a **router**, or a **digital camera**.

- ASICs are custom-designed chips optimized for a particular application or function.
- Unlike general-purpose processors, ASICs are tailored to execute specific tasks with maximum efficiency.
- They are commonly used in devices like digital voice recorders, high-efficiency video codecs, and cryptocurrency mining hardware.
- While ASICs offer high performance and low power consumption for their intended tasks, they lack flexibility and are costly to design and produce, making them suitable for high-volume production where the design cost can be amortized over many units

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ASIC

- A microchip designed to perform a specific or unique application. It is used as replacement to conventional general purpose logic chips.
- ASIC integrates several functions into a single chip and thereby reduces the system development cost.
- As a single chip, ASIC consumes very small area in the total system and thereby helps in the design of smaller systems with high capabilities/functionalities.
- ASICs can be pre-fabricated for a special application or it can be custom fabricated by using the components from a re-usable '*building block*' library of components for a particular customer application
- Fabrication of ASICs requires a non-refundable initial investment (Non Recurring Engineering (NRE) charges) for the process technology and configuration expenses

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RISC and CISC: Processor Architecture

- RISC is an abbreviation for Reduced Instruction Set Computer, while CISC is an abbreviation for Complex Instruction Set Computer. Out of these, the RISC processors have a comparatively smaller set of instructions along with few addressing modes. On the other hand, the CISC processors consist of a larger set of instructions along with multiple addressing modes.

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Processor Architecture: Reduced Instruction Set Computing (RISC)

- Uses fewer instructions, but they are very simple.
- Each instruction does only one small task.
- All instructions take the same short time to execute.
- Needs more lines of code to do something complex.
- Faster and more efficient for simple, repeated operations.
- RISC architectures utilize a small, highly optimized set of instructions, each designed to execute in a single clock cycle.
- This simplicity allows for faster instruction execution and easier pipelining, leading to improved performance and energy efficiency.
- RISC processors are prevalent in mobile devices and embedded systems, where power efficiency is crucial. Notable examples include ARM processors and the open-source RISC-V architecture, which has gained popularity due to its flexibility and lack of licensing fees

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Complex Instruction Set Computing (CISC)

- CISC architectures feature a comprehensive set of instructions, some capable of performing complex tasks in a single instruction.
- This design aims to reduce the number of instructions per program, potentially simplifying compiler design and improving code density
- However, individual instructions may take multiple clock cycles to execute, leading to increased complexity in decoding and execution.
- CISC processors, such as those based on the x86 architecture, are commonly found in desktop and server environments, where backward compatibility and rich instruction sets are advantageous

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ASIC,RISC CISC

RISC	CISC
Lesser no. of instructions	Greater no. of Instructions
Orthogonal Instruction Set (Allows each instruction to operate on any register and use any addressing mode)	Non Orthogonal Instruction Set (All instructions are not allowed to operate on any register and use any addressing mode. It is instruction specific)
Operations are performed on registers only, the only memory operations are load and store	Operations are performed on registers or memory depending on the instruction
Programmer needs to write more code to execute a task since the instructions are simpler ones	Instructions are like macros in C language. A programmer can achieve the desired functionality with a single instruction which in turn provides the effect of using more simpler single instructions in RISC
Less Silicon usage and pin count	More silicon usage since more additional decoder logic is required to implement the complex instruction decoding.

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Lecture 10

Comparison of microprocessor and microcontroller



Microprocessor

- ✓ A silicon chip representing a Central Processing Unit (CPU), which is capable of performing arithmetic as well as logical operations according to a pre-defined set of Instructions, which is specific to the manufacturer
- ✓ In general the CPU contains the Arithmetic and Logic Unit (ALU), Control Unit and Working registers
- ✓ Microprocessor is a dependant unit and it requires the combination of other hardware like Memory, Timer Unit, and Interrupt Controller etc for proper functioning.
- ✓ Intel claims the credit for developing the first Microprocessor unit Intel 4004, a 4 bit processor which was released in Nov 1971

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Microcontroller

- ✓ A highly integrated silicon chip containing a CPU, scratch pad RAM, Special and General purpose Register Arrays, On Chip ROM/FLASH memory for program storage, Timer and Interrupt control units and dedicated I/O ports
- ✓ Microcontrollers can be considered as a super set of Microprocessors
- ✓ Microcontroller can be general purpose (like Intel 8051, designed for generic applications and domains) or application specific (Like Automotive AVR from Atmel Corporation. Designed specifically for automotive applications)
- ✓ Since a microcontroller contains all the necessary functional blocks for independent working, they found greater place in the embedded domain in place of microprocessor. Microcontrollers are cheap, cost effective and are readily available in the market
- ✓ Texas Instruments TMS 1000 is considered as the world's first microcontroller

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Microcontroller

Microprocessor	Microcontroller
A silicon chip representing a Central Processing Unit (CPU), which is capable of performing arithmetic as well as logical operations according to a pre-defined set of Instructions	A microcontroller is a highly integrated chip that contains a CPU, scratch pad RAM, Special and General purpose Register Arrays, On Chip ROM/FLASH memory for program storage, Timer and Interrupt control units and dedicated I/O ports
It is a dependent unit. It requires the combination of other chips like Timers, Program and data memory chips, Interrupt controllers etc for functioning	It is a self contained unit and it doesn't require external Interrupt Controller, Timer, UART etc for its functioning
Most of the time general purpose in design and operation	Mostly application oriented or domain specific
Doesn't contain a built in I/O port. The I/O Port functionality needs to be implemented with the help of external Programmable Peripheral Interface Chips like 8255	Most of the processors contain multiple built-in I/O ports which can be operated as a single 8 or 16 or 32 bit Port or as individual port pins
Limited power saving options compared to microcontrollers	Includes lot of power saving features

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General Purpose Computers

- **Purpose:**

- Designed to perform a wide range of tasks, allowing users to install and run various applications.

- **Characteristics:**

- **Versatility:** Capable of handling diverse tasks, from word processing to gaming to software development.

- **User Programmability:** Users can install and run their own software.

- **High Resource Consumption:** Typically have more memory, processing power, and energy consumption compared to embedded systems.

- **Examples:** Personal computers, laptops, servers, and smartphones

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General Purpose Computers

Key Differences:

•Flexibility:

•Embedded systems are inflexible, optimized for a specific task, while general-purpose computers are versatile and can be used for many tasks.

•User Interaction:

•Embedded systems often have minimal user interaction, while general-purpose computers are designed for user interaction through software and interfaces.

•Resource Usage:

•Embedded systems are designed to be resource-efficient, while general-purpose computers typically require more resources like memory, processing power, and energy.

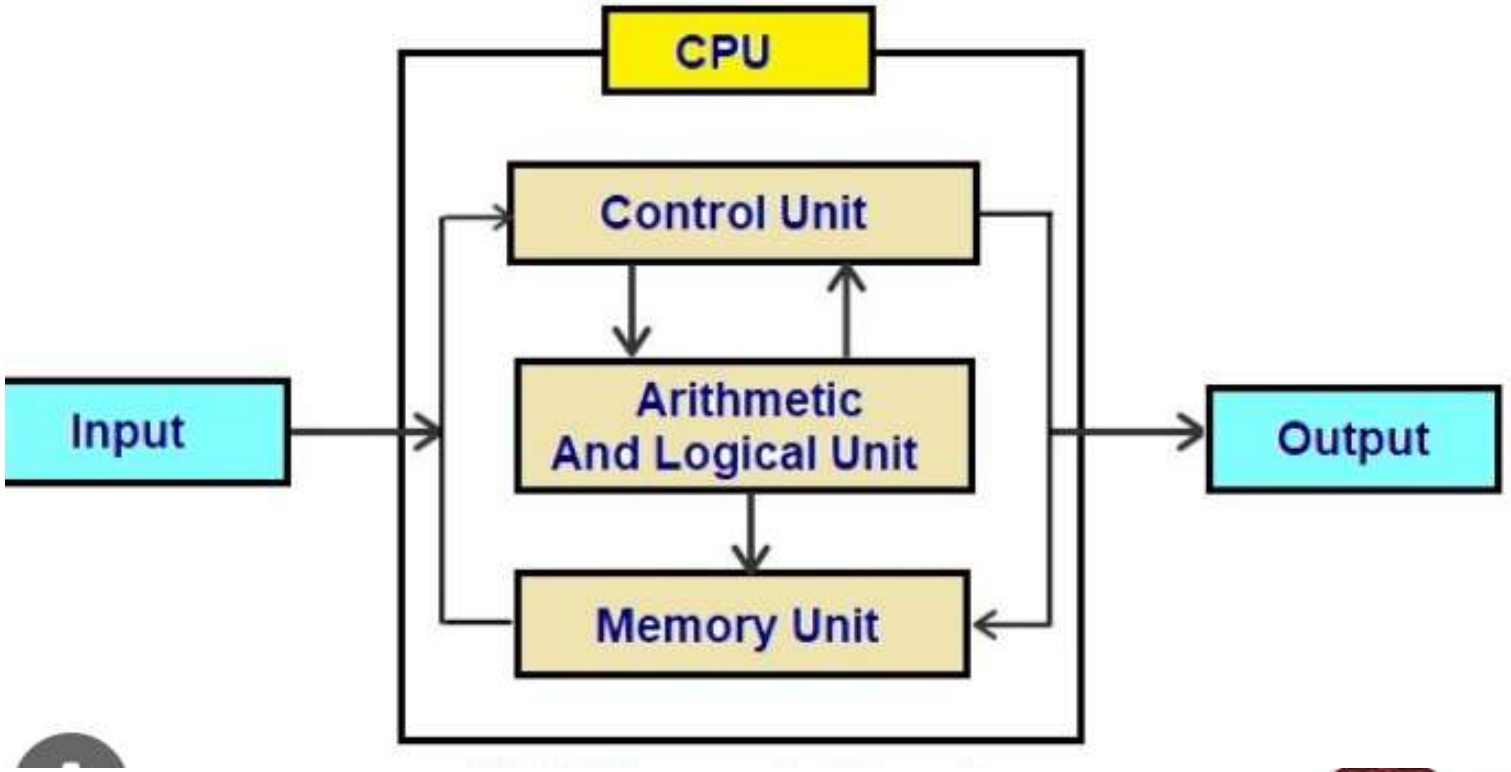
•Cost:

•Embedded systems are generally cheaper to design and build due to their specialized nature and limited resource usage

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General Purpose Computers

A general-purpose computer system's block diagram illustrates the interconnected functional units that enable processing data. These units include the input unit, the CPU (containing the Arithmetic Logic Unit and Control Unit), memory, and the output unit.



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Thank You



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